

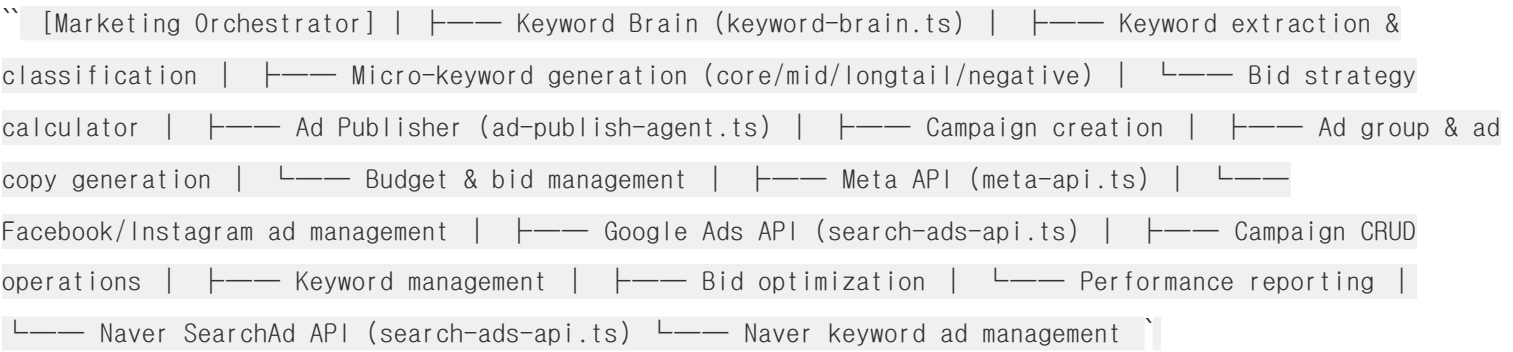
Google Ads API Tool - Design Documentation

Yeosonam (여소남) Marketing Automation Platform

1. Overview

Yeosonam is a B2B2C travel SaaS platform (<https://yeosonam.com>) that connects land operators to travel agencies and end customers in Korea. The Google Ads API integration is part of our marketing automation pipeline that automatically creates, manages, and optimizes keyword advertising campaigns for travel products.

2. System Architecture



3. Google Ads API Integration Design

3.1 Authentication Flow

- OAuth 2.0 with service account
- Developer token authentication
- Manager account (MCC: 313-217-4750) as parent for client accounts

3.2 Key Features	Feature	Description	API Endpoints Used			
	Campaign Management	Create/search/update campaigns	CampaignService		Ad Group Management	
		Create/manage ad groups with themed keywords	AdGroupService		Keyword Management	Add/update/bid on keywords at granular level
			AdGroupCriterionService		Bid Optimization	Micro-keyword bidding strategy with data-driven adjustments
			BiddingStrategyService		Performance Reporting	Daily/weekly performance data collection
			GoogleAdsService.search()		Budget Management	Campaign budget allocation and tracking
			CampaignBudgetService			

3.3 Micro-Keyword Bidding Strategy

Our core innovation is granular keyword bidding:

1. Keyword Classification Pipeline - Extract keywords from travel product data - Classify into: Core, Mid, Long-tail, Negative - Generate micro-variations (location + product + intent combinations)
2. Bid Strategy - Start with low CP bids for all micro-keywords - Collect 7-14 days of performance data - Apply automated bid adjustments based on CTR, conversion rate, ROAS - Gradually increase bids for high-performing

keywords – Pause or lower bids for underperforming keywords

3. Data Collection & Optimization Loop

Day 1–7: Low bid data collection phase
Day 8–14: Initial optimization (bid adjustments)
Day 15+: Continuous optimization based on accumulated data

3.4 Campaign Types Supported

- **Search Campaigns:** Primary focus for keyword advertising
- Automated ad copy generation from travel product data
- Responsive search ads with multiple headlines/descriptions

4. Data Flow

[Travel Product DB] → [Keyword Brain] → [Ad Publisher] ↓ [Google Ads API Service] ↓ [Campaign Creation] [Keyword Addition] [Bid Management] [Performance Tracking] ↓ [Analytics Dashboard]

5. Error Handling & Retry Logic

- Exponential backoff for API rate limits
- Graceful degradation: if Google Ads API fails, fall back to manual campaign management
- Logging all API interactions for audit and debugging
- Circuit breaker pattern for repeated failures

6. Security & Compliance

- OAuth 2.0 tokens stored encrypted in secret registry
- API keys managed through environment variables
- All API calls logged with request/response for audit
- Compliance with Google Ads API Terms & Conditions

7. Technology Stack

- **Runtime:** Node.js / Next.js (App Router)
- **Language:** TypeScript

- **Google Ads API Client Library:** google-ads-api` npm package (v21+)
- **OAuth:** Google OAuth 2.0 with refresh token rotation
- **Database:** Supabase (PostgreSQL) for campaign data storage
- **Secret Management:** Encrypted secret registry module

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